

A Note on the Apparent Polarity Status of English Bare *Much*

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1.

Pullum and Huddleston (2002, 827) point out that bare *much* is, in some clear cases, a Negative Polarity Item (NPI).¹ One of their examples is:

(1) I don't enjoy sailing much.

(2) *I enjoy sailing much.

They point out in addition that such clear cases are found in particular when bare *much* is "a post-verbal adjunct". I agree with their judgments. Some other cases that can be added are:

(3) You're *(not) around much these days.

(4) We *(don't) much like your attitude.

(5) Nobody/*Everybody listens to you much these days.

In this paper, I will try to understand why bare *much* acts like a NPI in (1)-(5). I will conclude that the NPI status of such instances bare *much* is a side effect of the licensing of a silent element, much as in Kayne (2021) on *long*. If this approach can be extended to other NPIs, the conclusion will be, in a more general way, that NPI-ness is not a primitive of the language faculty.²

2.

Jespersen (1970/1914, 106) took *few* to be an adjective on the basis of its having suffixal comparative and superlative forms *fewer* and *fewest*. Kayne (2002; 2005a; 2007) agreed with Jespersen's proposal, while adding the idea that the noun modified by adjectival *few* is generally silent. Thus in:

(6) John has few books.

few does not directly modify *books*. Rather it modifies a silent counterpart of the noun *number* (silence will be indicated by capitals):

(7) few NUMBER books

Bowers (1975, 542) took the related *little*, *many*, *much* to also be adjectives. The generalization from (7) to these is as in:

(8) little AMOUNT money

(9) many NUMBER books

(10) much AMOUNT money

The silent nouns in (7)-(10) must, as must silent elements in general, be licensed. The question is, how exactly, in one case or another. Let me begin by emphasizing the relative parallelism between *much money* and *a large amount of money*, with *much* now

¹Cf. Klima (1964, 284).

²Whether a generalization to positive polarity items (cf. Szabolcsi (2004)) is feasible is an open question.

close in this context to *large*, and with this parallelism leading to a revision of (10), namely to:

(11) A much AMOUNT money
in which a silent indefinite article A has been added, in a way that makes (11) resemble to a greater extent:

(12) A long TIME
which in Kayne (2021) underlies NPI *long*.³

At this point, we can bring in another point made explicitly by Pullum and Huddleston (2002, 826-7), to the effect that *much* becomes widely acceptable if deprived of its 'bare' status by the addition of a modifier, as in the following, for example, which contrast with (2):

(13) I enjoy sailing so/too/very much.

The core idea, now, is that the contrast between (2) and (1) is misleading. The unacceptability of (2) does not arise from its needing negation. It arises because (2) needs something extra. That something can be negation, or it can be a modifier of the sort shown in (13). Calling *much* an NPI is missing this broader generalization.

3.

As for the question why *much* needs something extra, I will try to show that that need arises from the presence of silent AMOUNT seen in (11), and more specifically from the licensing requirement for this AMOUNT.

Starting with *too*, part of the core idea is to relate the difference between *too much* and bare *much* to the following well-known facts:

(14) You bought too large a car.

(15) *You bought large a car.

which from the perspective of (11) transpose to:

(16) too much A AMOUNT

(17) *much A AMOUNT

with this pair then matching:

(18) You enjoy sailing too much.

(19) *You enjoy sailing much.

This matching holds because in (18), pre-silent indefinite article *too much* (as seen in (16), akin to (14)) licenses AMOUNT; in (19), bare *much* does not occur in that position (cf. (17), akin to (15)) and therefore cannot effect the licensing.

This initial attempt at an account requires saying that the licenser of AMOUNT must precede not only it, also the silent indefinite article; otherwise, bare *much* in (11) would be able to act as a licenser. But why should this particular precedence requirement hold? A possible answer, in the spirit of Kayne (2006), is that AMOUNT, in order to have its silence licensed, must itself raise past the (silent) indefinite article, and, furthermore, that the required raising of AMOUNT can only come about via AMOUNT being pied-piped by *too much*:

(20) A too much AMOUNT —>

[too much AMOUNT] A < too much AMOUNT >

³The similarity between *long* and *much* is noted by Pullum and Huddleston (2002, 827).

If so, *too much* does not directly license AMOUNT; rather, *too much* facilitates the licensing of AMOUNT by pied-piping it as in (20), with that pied-piping dependent in turn on the presence of *too*.⁴

4.

The clear contrast in acceptability between (18) and (19) is accentuated by the contrast, in turn, between (19) and the fully acceptable:

(21) You enjoy sailing a lot.

which reflects the fact that *a lot*, despite its interpretive similarity to *much*, shows no polarity behavior, and therefore must, from the present perspective, not be a modifier of AMOUNT in the way that *much* is.

A question arises now as to how best to express the fact that *a lot*, like *much*, feels close in interpretation to *a large amount*. The proposal has so far been that *much* is an adjectival modifier of silent AMOUNT.⁵ Taking *a lot* to also be an adjectival modifier (of any noun) seems implausible, given its form, with *lot* appearing to straightforwardly be a noun. A proposal that comes to mind, instead, is to take *a lot* to be as in:

(22) a LARGE lot

5.

Another fact about *much* is that its acceptability is sometimes dramatically improved by the addition of *very*:⁶

(23) We liked that film *(very) much.

From the preceding perspective, this suggests that *very much* can license AMOUNT in a way that bare *much* cannot, i.e. that the following must be available, parallel to the second line of (20):

(24) [very much AMOUNT] A < very much AMOUNT >

despite the fact that *very* often doesn't license pied-piping past the indefinite article as readily as *too*. This is seen in:

(25) ??You really bought very large a car.

though (25) is more acceptable than (15), repeated in essence here:

(26) *You really bought large a car.

and the following, with *of* added in colloquial fashion, seems even a bit more acceptable:

(27) ?You really bought very large of a car this year.

6.

The polarity sensitivity of bare *much* was illustrated in (3), repeated here:

⁴In the manner of Hendrick (1990).

⁵An open question is why adjectival *much* would not have a comparative **mucher* - for relevant discussion, see Kayne (2024), where it is noted that, if, along the lines of Barbiers's (2007, 869), *much* were to turn out to be a noun, the question would be answered.

⁶The extent to which this *very* is akin to Mandarin *hen* (cf. Huang (2015, sect. 2.2)) and/or to Cantonese *hóu* (cf. Matthews and Yip (1994, sect. 9.1.3)) needs to be determined.

(28) You're *(not) around much these days.

The contrast embedded in (28) has to my ear, somewhat surprisingly, a counterpart with *little*, although the contrast with *little* is less salient:

(29) You're around ?(very) little these days.

This suggests (at least) four things. First, *little* is, like *much*, accompanied by silent AMOUNT. Second, *very little* is parallel to *very much* in its ability to license AMOUNT. Third, bare *little* is able to license AMOUNT much more readily than bare *much*, presumably because *little*, unlike *much*, is associated with a syntactically present silent negation. And fourth, that silent negation is less strong as a pied-piper than overt negation.

7.

Negation interacts with *much* in another way:

(30) Nothing much is happening these days.

and related to this:

(31) I don't think that anything much is happening these days.

as well as:

(32) Not much of anything is happening these days.

in all of which *much* is c-commanded by negation.⁷

If negation is entirely absent, we unsurprisingly have:

(33) *Something much/*Everything much is happening these days.

That negation must end up c-commanding *much* to license it is shown by the following contrast:

(34) I don't think that much of anything is happening these days.

(35) *I think that much of nothing is happening these days.

8.

Returning to the basic contrast between bare *much* and modified *much* illustrated again in:

(36) I enjoy sailing *(so) much.

we can note a partially similar contrast in:

(37) *I see that you've given much a great deal of thought.

vs.

(38) ?I see that you've given much else a great deal of thought.

which suggests, given all of the preceding discussion, and in particular (24), that *else* in

(38) can to some extent act as a pied-piper of *much*:

(39) [much else AMOUNT] A <much else AMOUNT > NOUN

with in this case a silent noun present that may be STUFF or THING.

9.

On the other hand, bare *much* is perfectly possible, apparently in the absence of any licenser, in:

(40) Mary is much more/less intelligent than John.

⁷*Not much* may have moved into Spec, *of* in the manner of Kayne (1993, sect. 1.2; 1994, sects. 8.1, 8.5).

- (41) They're much too slow.
 These, however, contrast with:
 (42) *Mary is more/less intelligent than John by much.
 (43) *They're too slow by much.

As the following show, it's not that *much* is incompatible with this kind of *by*-phrase:

- (44) Mary is not more/less intelligent than John by much.
 (45) They're not too slow by much.

Clearly, *not* is the licenser of *much* in (44) and (45), with pied-piping as in:

- (46) by [not much AMOUNT] A < not much AMOUNT >

and with *not* subsequently raising further.⁸

As for (40) (and perhaps, too, for (41)), it may be relevant that in some languages, e.g. French, comparatives license a so-called 'expletive negation',⁹ shown in:

- (47) Marie est plus intelligente que Jean ne l'est. ('M is more intelligent than J neg. it is')

Possibly, then, (40) (but not (42)) allows *much* to be licensed by a silent raised English counterpart NE of the *ne* of (47). Alternatively, in both (40) and (41), *much* is being licensed by *more/less/too*; if so, then in (40) and (41) *much* will have raised (perhaps via remnant movement) from a position below *more/less/too*.

10.

Pullum and Huddleston (2002, 826) note other kinds of examples, "more characteristic of written English, literary contexts, formal style, and the usage of older speakers", in which *much* seems to lack a licenser. In my English, one such example would be:

- (48) The chemists have been much favored by the current administration.

which contrasts somewhat with:

- (49) ?The current administration has much favored the chemists.

and more sharply with:

- (50) *The current administration has favored the chemists much.
 (51) *The chemists have been favored by the current administration much.

with (50)/(51) in turn sharply different from:

- (52) The current administration has favored the chemists a lot/a great deal.
 (53) The chemists have been favored by the current administration a lot/a great deal.

Why bare *much* is allowed in pre-participial position in the relevant registers of English in (48) and (49), yet disallowed post-verbally in (50) and (51) is not clear to me, though it feels similar to facts concerning fairly archaic *thrice*:

- (54) ?They were thrice warned, yet did nothing.
 (55) *They were warned thrice, yet did nothing.

11.

Pullum and Huddleston (2002, 827) point out that to a certain extent bare *many* has the behavior of bare *much*. They give:

- (56) We saw many flowers.

⁸As in Collins and Postal (2014); cf. also Chomsky (1973, 242).

⁹Cf. Delfitto (2020).

as “very slightly literary”, as opposed to the ordinary:

(57) We didn’t see many flowers.

And they note that this polarity-like effect is strong in informal English. Slightly adapting an example of theirs, my informal English has:

(58) I ate a lot of cookies, that’s why I’m not hungry.

(59) *I ate many cookies, that’s why I’m not hungry.

I don’t have an account of why only informal English shows this strong effect.

However, for informal English, the above set of proposals for *much* appears to carry over. Just as adding a degree modifier like *too* changes the judgments on *much*, as in (13), so, too, does adding a degree modifier change the status of (59):

(60) I ate too many cookies, that why I’m not hungry.

Parallel to (39), then, in which *much* is a modifier of AMOUNT, *many* must in general be a modifier of NUMBER.¹⁰ And by extension from (20), we have in (60) pied-piping:

(61) [too many NUMBER] A < too many NUMBER > cookies

in which *too (many)* is the pied-piper and A a silent counterpart of the indefinite article seen in:

(62) You’ve eaten too large a number of cookies.

In informal English, (59) is not possible because there is no pied-piper like *too* present.

Furthermore, just as in (29) with *little*, we find that (59) vs. (60) has, surprisingly, a counterpart with *few*, though, again, the effect is less strong than with *many*.¹¹

(63) ?It took them few days to finish their paper.

(64) It took them very few days to finish their paper.

As with *little*, we can draw a number of conclusions. First, *few*, like *many*, is accompanied by silent NUMBER. Second, *very few* is parallel to *very many* (and *too many*) in its ability to license NUMBER. Third, bare *few* is able to license NUMBER more readily than bare *many*, presumably because *few*, unlike *many*, is itself associated with a syntactically present silent negation. And fourth, that silent negation is less strong as a pied-piper than overt negation.

12.

The apparent negative polarity behavior seen with *much* and *many* is a side effect of a licensing requirement holding of the silent AMOUNT or NUMBER that they cooccur with, much as in Kayne (2021) on *long*. An effect of this licensing requirement is seen, too, with *little* and *few*, even though they are not polarity items. The overall conclusion is that negative polarity is not a primitive of the language faculty, at least not in the kind

¹⁰Or perhaps QUANTITY - cf. Kayne and Pollock (2025).

¹¹With an overt indefinite article, the marginality effect disappears:

i) a little money; a few dollar bills

for reasons that need to be worked out. *A little* and *a few* are not entirely parallel:

ii) *another little money

iii) another few dollar bills

Nor are *much* and *many*, in the presence of an indefinite article:

iv) *a great/good much money

v) a great/good many dollar bills

in a way that recalls the stronger polarity behavior of *much*, as compared with *many*.

of cases discussed. Silent elements and their licensing requirements are more fundamental.

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